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Methodology

To evaluate the relative complexity of each process, we introduce a novel **Processing Intensity Score (PIS)** that integrates three dimensions: (i) a lexical-based Word Complexity Score, (ii) an Embedding Dispersion Score measuring semantic variability, and (iii) a Cluster Transition Difficulty Score reflecting how distant a process is from the global cooking-technique space. This normalized, interpretable score enables ranking individual processes as well as comparing clusters on an objective processing scale. Finally, we propose the **CHEF Classification System**, a five-level scale ranging from *minimal* to *highly technical*, which categorizes each process and cluster based on its computed intensity.

The dataset used in this project comes from **RecipeDB (Cosylab)**, a large curated repository of global recipes. It combines structured metadata, ingredients, processes, and cooking instructions into a unified format. Only the most relevant columns were used for analysis:

- **Recipe ID** – Unique identifier for each recipe.
- **Recipe Title** – Name of the dish.
- **Total Time** – Total preparation and cooking time.
- **Ingredients** – Cleaned list of ingredients used.
- **Instructions / Steps** – Full cooking steps used for extracting processes and training embeddings.
- **Processes** – Extracted cooking actions (e.g., *heat*, *stir*, *chop*, *simmer*).
- **Process Count** – Total number of operations per recipe
- **Region / Sub-Region / Continent** – Geographic origin of the recipe for cultural comparisons.
- **Diet Labels** – Vegetarian, vegan, pescetarian, etc., for nutritional/dietary grouping.
- **Basic Nutrients** – Energy, fat, carbs, protein

- **place** — 55,651

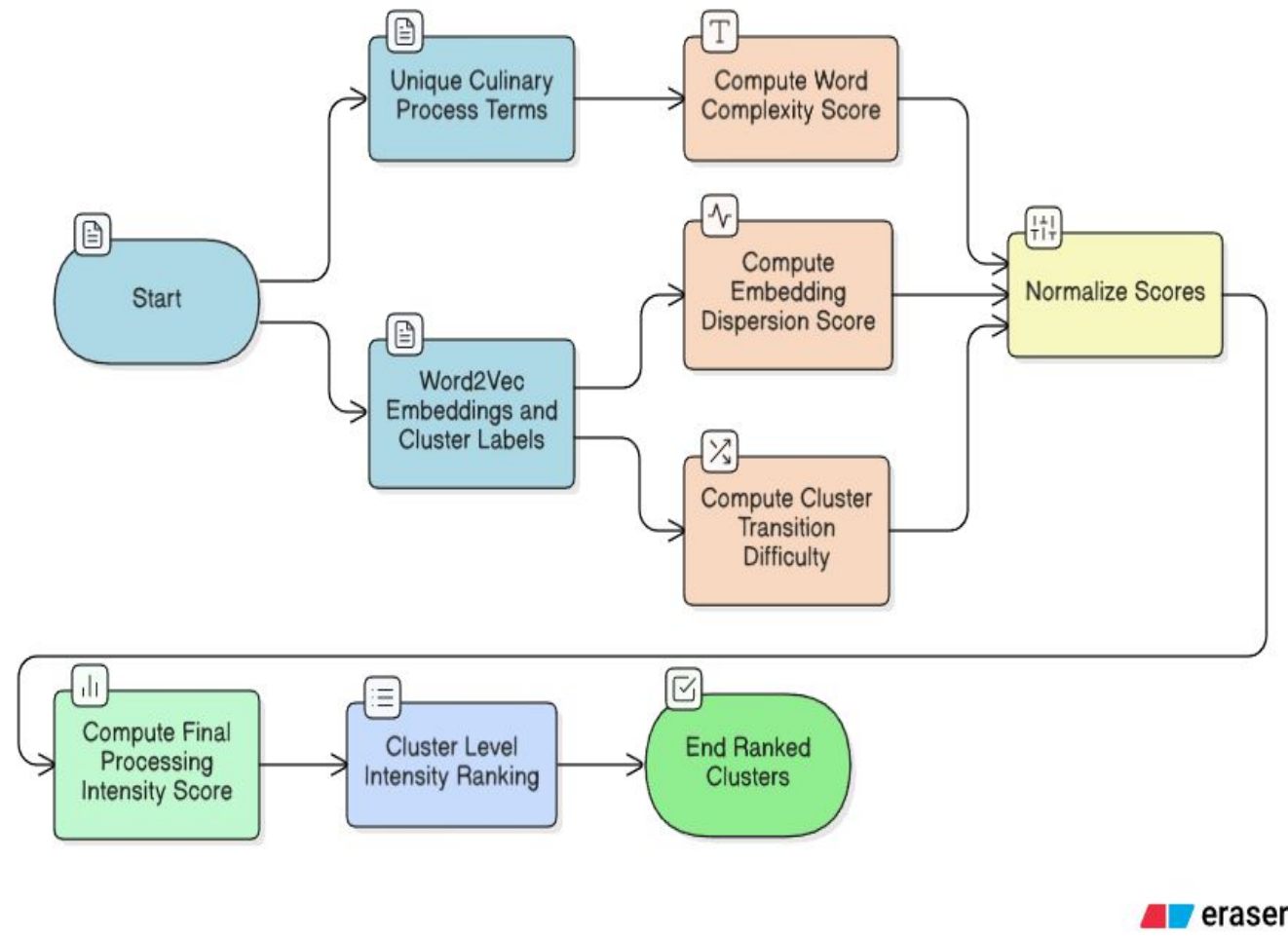
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1. **K-Means:** Groups process embeddings into a fixed number of clusters by minimizing within-cluster distance.
2. **Agglomerative Clustering:** Builds clusters hierarchically by repeatedly merging the closest groups.
3. **DBSCAN:** Forms clusters based on dense regions in the data and marks low-density points as noise.
4. **Gaussian Mixture Model (GMM):** Creates soft, probabilistic clusters assuming the data is generated from multiple Gaussian distributions.

[illegible]

Cluster_Intensity(k) = average(PIS of all processes in cluster k)

Work Flow



RESULTS

Process	PIS
add	0.001
set	0.013
mix	0.018
put	0.020
take	0.070

References

da Costa Louzada, M. L. & Gabe, K. T. (2025). *Nova food classification system: a contribution from Brazilian epidemiology*. Revista Brasileira de Epidemiologia, 28, e250027. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12129239/>

Complete work details

- Food Pairing Analysis